

Frontend Layer Architecture - StudyWithUs

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This document defines the frontend layer stack and their runtime interaction for the React + TypeScript application.

1. Layer Stack Diagram

Layered Frontend Architecture

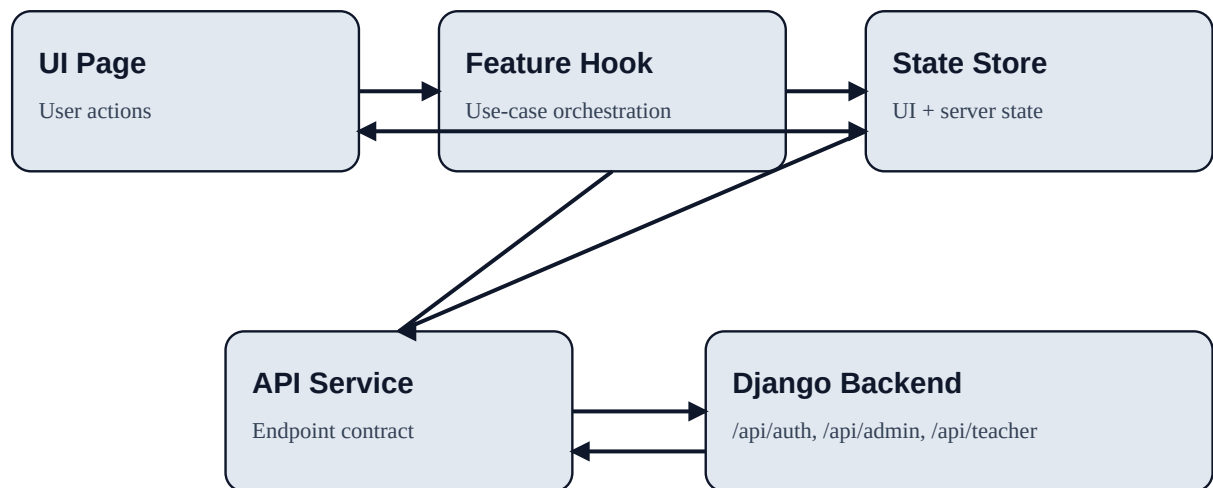


Flow direction: User events go downward to data/infrastructure; results propagate upward to UI.

2. Runtime Interaction Diagram

The following diagram shows how UI events flow through feature logic, state, and API calls.

Layer Interaction (Runtime)

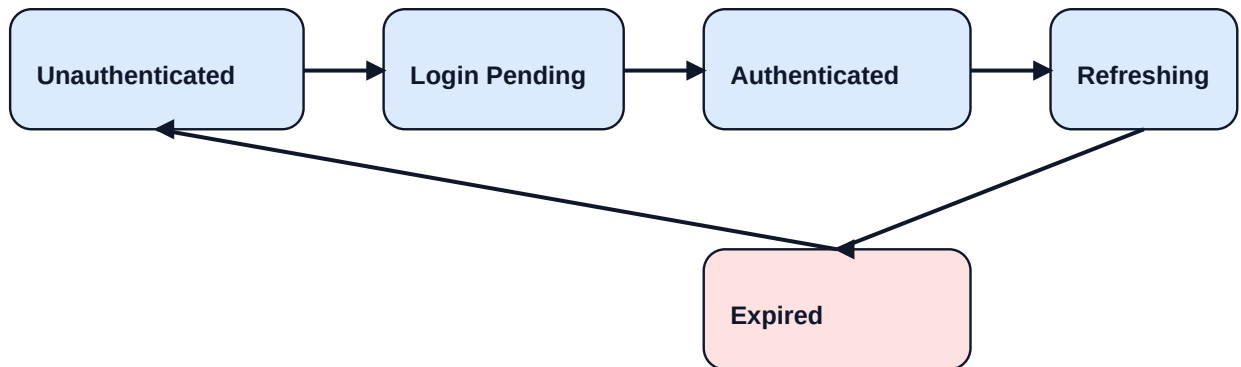


1) UI triggers feature action 2) Feature updates state and calls API 3) Response updates state 4) State rerenders UI

3. Session State Flow

Authentication-related state transitions used by route guards and protected pages.

Session State Flow



Trigger events: login submit, token expiry, refresh success/failure, explicit logout.

4. Layer Responsibilities

- Presentation Layer: route tree, role guards, page composition, reusable UI blocks.
- Feature Layer: use-case specific logic (auth, class management, schedule, attendance, forum, exams, PFE).
- Domain Layer: shared typed models and rule translation from backend payloads.
- Data Access Layer: endpoint methods and cache invalidation orchestration.
- Infrastructure Layer: HTTP client, token persistence, environment configuration.